

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)
Course Name: Applied Physics Course Code: PHY-111 Examination: 1st Periodical
Scheme: CBCS Branch: CE/ECE/CSE Semester: 1st
Faculty Name: Dr. Neelam Guleria

Course Outcomes (COs): Upon successful completion of the course, students will be able to:

CO1: Recognize the importance of Physics in describing different physical phenomenon.

CO2: Comprehend the Special theory of relativity by examining inertial & non-inertial frames, Michelson-Morley experiment, Galilean & Lorentz transformations.

CO3: Demonstrate the concept of harmonic oscillators by studying simple harmonic, damped harmonic and forced harmonic oscillations.

CO4: Illustrate the concept of LASER, Optical Fibres and X-rays in various fields.

CO5: Appraise the concept of electromagnetic waves and Superconductivity in the field of Engineering.

CO6: Recognize the inadequacy of classical mechanics for certain physical problems and thus find the solution of these problems using Quantum Mechanics.

Question Paper for 1st Periodical Examinations (October, 2023)

Time: 2 Hours

Max. Marks: 50

Note: Attempt all questions.

Q. No.	Sub Q. No.	CO(s)	Question	Max. Marks
Q1.	(i)	CO2	Is earth an inertial or non inertial frame of reference? Explain with reason.	2
	(ii)	CO2	Calculate heat energy produced when 1 kg mass is completely converted into heat energy.	2
	(iii)	CO2	Calculate rest mass & relativistic mass of photon of energy 5eV.	2
	(iv)	CO2	State postulates of Special theory of relativity.	2
	(v)	CO4	Calculate energy difference (in eV) between two levels of Ne atoms of He-Ne laser which results in the emission of a light of wavelength 632.8 nm.	2
Q2.		CO1, CO2	Explain in detail Michelson-Morley experiment. How is the negative result of the experiment interpreted?	10
Q3.		CO2	Deduce the expression for length contraction and time dilation from Lorentz transformations.	10
Q4.		CO1, CO4	Stimulated absorption, Stimulated emission and Spontaneous emission. Derive the relation between Einstein's coefficients?	10
Q5.		CO1, CO4	Explain the principle, construction, working & output of Ruby Laser. Or Explain the principle, construction, working & output of He-Ne Laser.	10